

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 30 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 30 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 30 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 31 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 31 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 30 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 32 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 30 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 30 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 30 trials
Total training paths	16 · 454 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/1_Alán/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports/SingleHand-Experiment/LeftHand

Sensor selection

Hands	left
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	88 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4752
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=363 · test=91

Augmentation (training only)

Enabled	no
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Voting Soft (SVM+RF+KNN+LogReg)	0.9505	0.0471	0.9670
Stacking (SVM+RF+KNN+LogReg)	0.9725	0.0300	0.9670
SVM (Linear)	0.9586	0.0332	0.9451
Random Forest	0.9753	0.0378	0.9451
Logistic Regression	0.9450	0.0441	0.9451
SVM (RBF)	0.9531	0.0392	0.9341
LDA	0.8815	0.0494	0.9121
KNN	0.7464	0.0485	0.6593

Per-class hold-out F1

[illegible]

